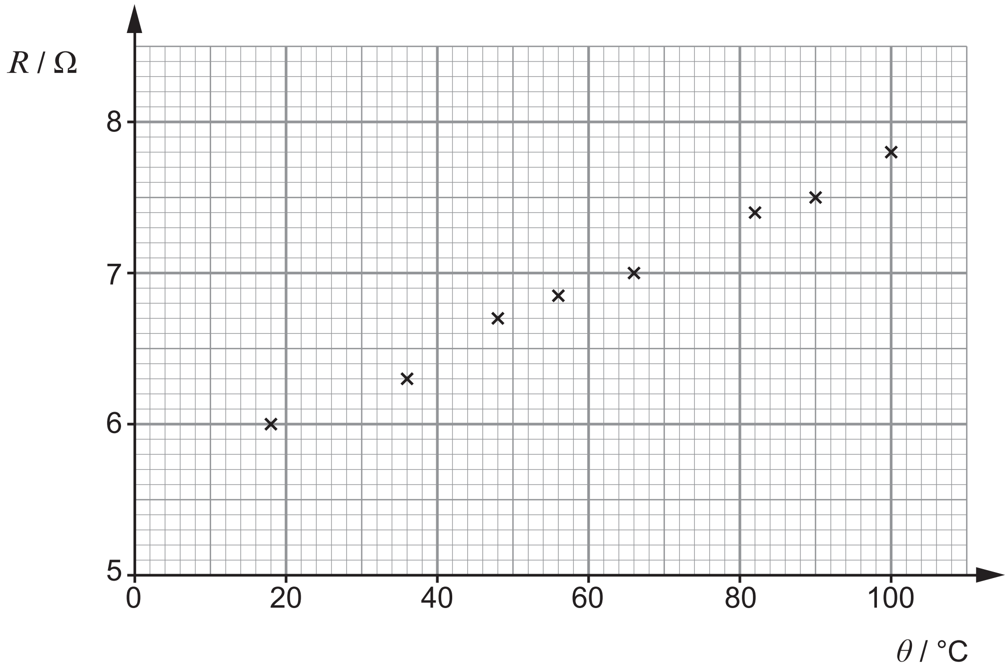
3. A student slowly heated a coil of insulated copper wire. He took readings of its temperature, θ, and resistance, R, at intervals. The readings are plotted below.



(a) Bearing in mind the range of temperatures, suggest how the student heated the coil, and how he could have extended the range of temperatures down to just above 0 °C. [2]

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(b) Textbooks state that the resistance, R, of a metal wire is related to its celsius temperature, θ, by an equation of the form:

$$R = R_0 \alpha \theta + R_0$$

in which R_0 and α are positive constants.

(i) Explain how the readings plotted support the equation. [3]

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Examiner only

(ii) Determine from the graph the values, with units, of:

I. R_0; [1]

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II. α. [3]

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